

TUTORIAL – 2: TEST PLAN & REQUIREMENT TRACEABILITY MATRIX (RTM)

Aim:

To understand a professional Test Plan, Requirement Traceability Matrix (RTM), Testing Scope Document, and Testing Strategy using a sample application.

Description

A Test Plan is a document that explains how software testing will be performed. It includes the testing scope, objectives, resources, schedule, and testing methods.

A Requirement Traceability Matrix (RTM) is a table that maps each software requirement to its corresponding test cases. It ensures that every requirement is tested and nothing is missed.

Testing Scope defines which features are included and excluded from testing.

Testing Strategy describes the overall approach followed during software testing.

Problem Statement

Prepare a Test Plan and Requirement Traceability Matrix (RTM) for a **Student Management System**. Design test cases for the Login Module and verify whether all functional requirements are covered through testing.

Implementation Details

The Student Management System allows users to:

- User Login
- Add Student
- Update Student Details
- Delete Student
- Search Student
- Logout

Testing is performed on the Login Module.

Parameter Details

Parameter	Details
Project Name	Student Management System
Application/Software	Desktop/Web Application
Module Name	Login Module
SDLC Model	Waterfall Model
Programming Language	Java/Python/PHP

IDE/Tool Used	Eclipse / VS Code / PyCharm
Input/Test Data	Username: admin Password: admin123

Source Code

(Dummy Login Program in Python)

```
username = input("Enter Username: ")  
password = input("Enter Password: ")
```

```
if username == "admin" and password == "admin123":  
    print("Login Successful")  
else:  
    print("Invalid Username or Password")
```

Requirement Traceability Matrix (RTM)

Requirement ID	Requirement Description	Test Case ID	Status
R001	Login with valid credentials	TC001	Pass
R002	Reject invalid username	TC002	Pass
R003	Reject invalid password	TC003	Pass
R004	Display error message	TC004	Pass

Test Case Detailed

Field	Details
Test Case ID	TC001
Requirement ID	R001
Module Name	Login
Feature Under Test	User Authentication

Test Scenario	Login using valid username and password
Test Objective	Verify successful login
Test Type	System Testing
Test Technique	Black Box Testing
Priority	High
Severity	Critical
Preconditions	Application is installed and opened
Postconditions	User enters Home Page
Test Environment	Windows 10
Browser/OS	Google Chrome / Windows 10
Test Data	admin / admin123
Expected Result	Login Successful
Actual Result	Login Successful
Status	Pass
Executed By	Student Name
Execution Date	DD/MM/YYYY

Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open Application	-	Login Page opens	Login Page opens	Pass
2	Enter Username	admin	Username accepted	Username accepted	Pass

3	Enter Password	admin123	Password accepted	Password accepted	Pass
4	Click Login Button	Click	Home Page displayed	Home Page displayed	Pass

Testing Activities Performed

Sr. No.	SDLC Phase	STLC Phase	Testing Activity	Deliverable/Artifact
1	Requirement Analysis	Requirement Analysis	Study Requirements	SRS Document
2	System Design	Test Planning	Prepare Test Plan	Test Plan
3	Development	Test Case Design	Create Test Cases	Test Case Document
4	Coding	Test Execution	Execute Test Cases	Test Report
5	Integration	Defect Reporting	Report Bugs	Bug Report
6	Testing	Regression Testing	Re-test after Fix	Updated Report
7	Deployment	Test Closure	Final Testing Summary	Test Closure Report

Observation

Sr. No.	Observation	Remarks
1	Login works with valid credentials	Pass
2	Invalid username rejected	Pass
3	Invalid password rejected	Pass

4	Error message displayed correctly	Pass
5	Login page loads successfully	Pass

Output

Enter Username : admin

Enter Password : admin123

Login Successful

Solve the Questionnaire

1. What is the primary purpose of a Test Plan?

A) Coding Standards

B) Testing Management

C) Database Design

D) Deployment

Answer: B) Testing Management

2. What does RTM stand for?

A) Requirement Technical Matrix

B) Requirement Traceability Matrix

C) Resource Tracking Matrix

D) Requirement Test Matrix

Answer: B) Requirement Traceability Matrix

3. What is the main purpose of RTM?

A) To create source code

B) To map requirements with test cases

C) To design the database

D) To deploy software

Answer: B) To map requirements with test cases

4. Which document defines functionalities that are included and excluded from testing?

A) User Manual

B) Testing Scope Document

C) Source Code

D) Design Specification

Answer: B) Testing Scope Document

5. Which document defines the overall testing methodology and approach?

A) Bug Report

B) RTM

C) Testing Strategy

D) User Guide

Answer: C) Testing Strategy

Conclusion / Outcome

- Understood the purpose and structure of a **Test Plan**.
- Learned how to prepare a **Requirement Traceability Matrix (RTM)**.
- Designed and executed test cases for the Login Module.
- Verified that all software requirements were covered through testing.
- Gained practical knowledge of the software testing process and documentation.